

Network Troubleshooting, Management, Monitoring, and Optimization

Task 1 :- Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

ANS :-

```
Microsoft Windows [Version 10.0.26200.9445]
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C:\Users\asvin>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    IPv6 Address. . . . . : 2409:40c1:4025:2267:b779:f9f8:c26c:879b
    Temporary IPv6 Address. . . . . : 2409:40c1:4025:2267:8deb:cda3:d59b:3c6f
    Link-local IPv6 Address . . . . . : fe80::9c9:38c:7b17:d61c%2
    IPv4 Address. . . . . : 10.86.106.47
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1c3a:5cff:fea7:37a8%2
                                10.86.106.3

C:\Users\asvin>
```

IPv4 Address. : 10.86.106.47

Subnet Mask : 255.255.255.0

Default Gateway : fe80::1c3a:5cff:fea7:37a8%2 10.86.106.3

DNS Servers. : 10.86.106.3

Task 2 :- Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.

ANS :-

Comparison & Meaning :

State	Output Status	What It Means
Disconnected	Request timed out / Destination host unreachable	No connection. Data cannot leave your computer or find a path to Google's server.
Reconnected	Reply from 142.250.x.x: bytes=32 time=20ms	Connected! Your computer successfully sent data packets to Google and received them back.

Task 3 :- Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.

ANS :-

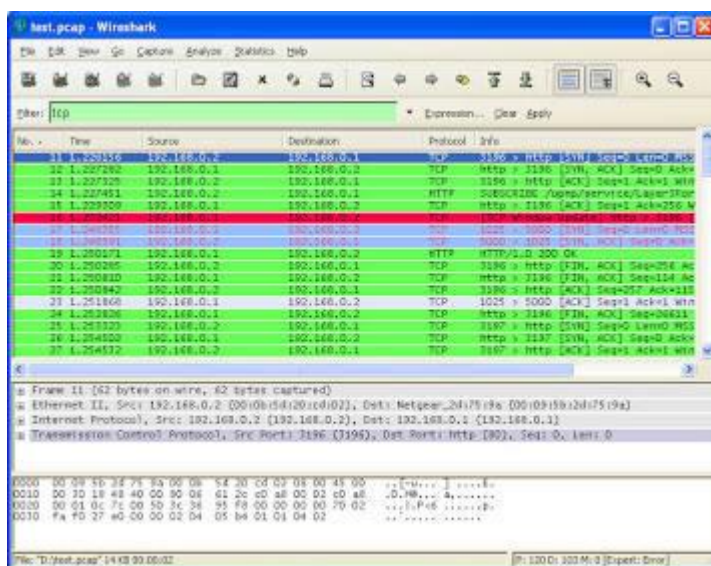
File Options View				
Processes Performance App history Startup Users Details Services				
Name	38% CPU	63% Memory	19% Disk	0% Network
System interrupts	0.1%	0 MB	0 MB/s	0 Mbps
> Task Manager	4.5%	11.1 MB	0 MB/s	0 Mbps
COM Surrogate	0%	0.9 MB	0 MB/s	0 Mbps
COM Surrogate	0%	0.9 MB	0 MB/s	0 Mbps
WMI Provider Host	0%	1.4 MB	0 MB/s	0 Mbps
Groove Music	0%	26.5 MB	0 MB/s	0 Mbps
Application Frame Host	0%	5.1 MB	0 MB/s	0 Mbps
> Service Host: Network Connecti...	0%	0.8 MB	0 MB/s	0 Mbps
WMI Provider Host	0%	1.4 MB	0 MB/s	0 Mbps
WMI Reverse Performance Ada...	0%	1.1 MB	0 MB/s	0 Mbps
Google Chrome	9.1%	195.7 MB	0.1 MB/s	0 Mbps
> Service Host: Unistack Service G...	0%	2.0 MB	0 MB/s	0 Mbps
> Service Host: Security Center	0%	1.0 MB	0 MB/s	0 Mbps
> Service Host: Delivery Optimizat...	0%	1.2 MB	0 MB/s	0 Mbps
Google Chrome	0%	8.0 MB	0 MB/s	0 Mbps
Fewer details				End task

Typical Example Results :

Metric	Recorded Value
Top Process	Google Chrome (or msedge.exe)
Peak Usage	15 – 35 Mbps (spikes high when video loads, then drops)

Task 4 :- Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets.

ANS :-



3 Things You Can See :

1. IP Addresses : The source (your PC IP) and destination (Flipkart/Instagram server IP).
2. Website Domain Name (Server Name) : The domain name requested during the secure handshake (SNI).
3. Packet Size & Time : Exact timestamp and size (in bytes) of each data chunk transferred.

Task 5 :- Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

ANS :-

1. AI Checklist
 1. Restart the router.
 2. Use 5 GHz Wi-Fi.
 3. Move closer to the router.
 4. Stop other downloads.
 5. Update the router firmware.
 6. Use Ethernet for gaming.

★ First, I would try: Using an Ethernet cable.

